

Sample and Population

Econ 140 Fall 2025, Section 2

Jonathan Old

[Syllabus/OH](#) [bCourses](#) [Website](#) [Feedback form \(Always open\)](#) [Datahub](#)

Roadmap

1. Getting Started & Housekeeping
2. Time for your questions
3. Intro to today's section
4. The Law of Large Numbers (LLN)
5. The Central Limit Theorem (CLT)

Getting Started & Housekeeping

Reminder about office hours

- Email: jonathan_old@berkeley.edu

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- Can use [Econ Department tutoring services \(See here\)](#).

Problem Set 1

- Due: Tuesday September 16 at Midnight, via Gradescope.
- Deliverable: A short (2–3 page) report or slide deck.
- Teamwork encouraged

Your questions

Any questions?

...Remember – this is a safe space! Every question is useful!

Intro to today's section

How can learn from data?

- We constantly use "samples" of data to learn about things we care about
 - Learning about what is the best restaurant in Berkeley
 - Finding out the best sports team in a league
 - We will have two midterms and a final exam to learn about how well you studied for Econ 140
- Today we focus on the process of sampling, and how we can guarantee that the sample we draw allows us to learn something about the real world, and not just the data.

What can go wrong?

- Suppose you want to know what the best restaurant in Berkeley is. For that, you decide to estimate the quality of each restaurant.
- You decide to use the following sampling strategies:
 - Get all restaurants from Google Maps and use the average rating
 - Ask your friends what their favorite restaurant is, quality is number of mentions
 - Get a list of all restaurants in Berkeley and visit all of them once to rate them
- Take 5 minutes with your group and discuss what the problems are with each strategy.



Sample Size Matters

... and many other things matter as well: Data quality, measurement, comparability of measurements, etc.

How parachute companies feel after realizing that it's impossible for them to get bad reviews



Selection Matters

- The Law of Large Numbers (LLN) and Central Limit Theorem (CLT) are two fundamental results in statistics that describe the behavior of sample averages as the sample size increases.
- LLN states that as the sample size increases, the sample average converges to the population mean.
- CLT states that as the number of samples increases, the distribution of the sample average approaches a normal distribution, regardless of the population's distribution.
- Let us work it out together on the worksheet.

The Law of Large Numbers (LLN)

The Law of Large Numbers (LLN)

- Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with mean μ and finite variance σ^2 .
- The sample mean is given by:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

- The LLN states that as n approaches infinity, the sample mean $\bar{X}_n$ converges in probability to the population mean μ :

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| < \epsilon) = 1$$

for any $\epsilon > 0$.

- Intuition: As the sample size grows, the average of the sample gets closer to the true population mean. "The more data we collect, the better we approximate the truth."

LLN and restaurant quality

- What is the population mean? – It depends on our question
- Suppose we want to estimate: How much do I like cheeseboard pizza?
 - Population: All possible times I could eat cheeseboard pizza
 - Sample: The times I have eaten at cheeseboard pizza
 - Sample mean: Average rating I give to cheeseboard pizza
 - As I visit cheeseboard more and more, my average rating will converge to my "true rating" of cheeseboard pizza
 - But if I only visit once, it might have been a particularly good or bad day
 - What would be a violation of (1) independence, (2) identical distribution?

LLN and restaurant quality II

- Suppose we want to estimate the average quality of all restaurants in Berkeley
 - Population: All restaurants in Berkeley
 - Sample: The restaurants I have visited in Berkeley
 - Sample mean: Average rating I give to the restaurants I have visited
 - As I visit more and more restaurants, my average rating will converge to the "true average quality" of all restaurants in Berkeley
 - But if I only visit a few, there may be outliers, or I may have a biased sample (only South Berkeley)
 - What would be a violation of (1) independence, (2) identical distribution?

We will code on **Datahub**.

The Central Limit Theorem (CLT)

The Central Limit Theorem (CLT)

- Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables with mean μ and finite variance σ^2 .
- The sample mean is given by $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- The CLT states that as n approaches infinity, the distribution of the sample mean $\bar{X}_n$ approaches a normal distribution with mean μ and variance $\frac{\sigma^2}{n}$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

where $\xrightarrow{d}$ denotes convergence in distribution.

- Intuition: Regardless of the original distribution of the data, the distribution of the sample mean will be approximately normal for large sample sizes. The average of many random variables tends to be normally distributed.

Explanation using Settlers of Catan



Explanation using Settlers of Catan

- The distribution of a single die roll is uniform, not normal
- For two dice, the distribution of the sum is triangular
- For three dice, it starts to look more like a normal distribution
- As we increase the number of dice, the distribution of the sum approaches a normal distribution
- This is the CLT in action: The sum (or average) of many independent random variables tends to be normally distributed, regardless of the original distribution of the variables.

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